```
%pip install pandas scikit-learn matplotlib
```

[3] ✓ 1.2s Python

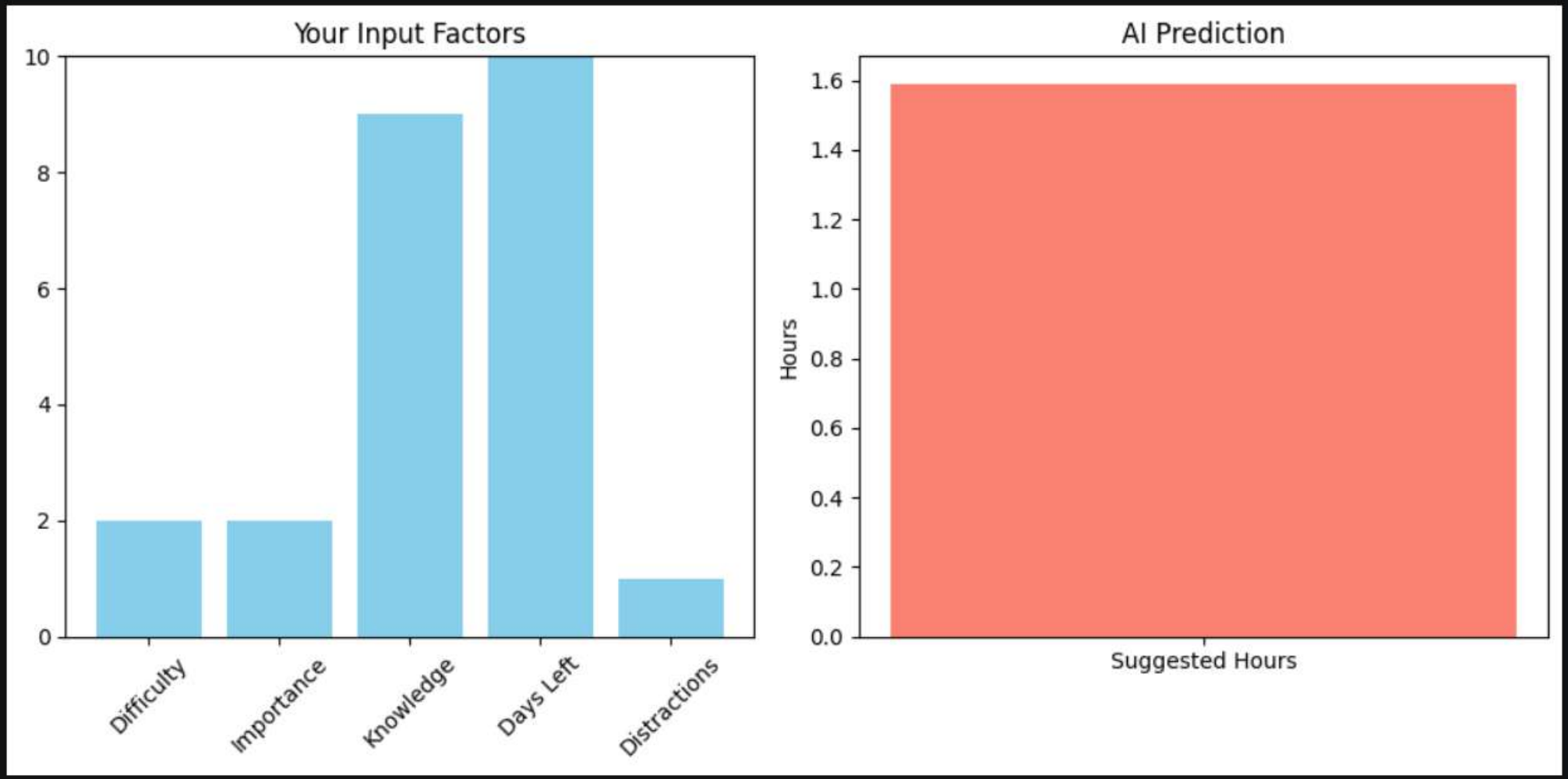
```
Defaulting to user installation because normal site-packages is not writeable
Requirement already satisfied: pandas in c:\users\sweet\appdata\local\packages\pythonsoftwarefoundation.python.3.12_qbz5n2kfra8p0\loc
Requirement already satisfied: scikit-learn in c:\users\sweet\appdata\local\packages\pythonsoftwarefoundation.python.3.12_qbz5n2kfra8
Requirement already satisfied: matplotlib in c:\users\sweet\appdata\local\packages\pythonsoftwarefoundation.python.3.12_qbz5n2kfra8p0
Requirement already satisfied: numpy>=1.26.0 in c:\users\sweet\appdata\local\packages\pythonsoftwarefoundation.python.3.12_qbz5n2kfra
Requirement already satisfied: python-dateutil>=2.8.2 in c:\users\sweet\appdata\local\packages\pythonsoftwarefoundation.python.3.12_q
Requirement already satisfied: tzdata in c:\users\sweet\appdata\local\packages\pythonsoftwarefoundation.python.3.12_qbz5n2kfra8p0\loc
Requirement already satisfied: scipy>=1.10.0 in c:\users\sweet\appdata\local\packages\pythonsoftwarefoundation.python.3.12_qbz5n2kfra
Requirement already satisfied: joblib>=1.3.0 in c:\users\sweet\appdata\local\packages\pythonsoftwarefoundation.python.3.12_qbz5n2kfra
Requirement already satisfied: threadpoolctl>=3.2.0 in c:\users\sweet\appdata\local\packages\pythonsoftwarefoundation.python.3.12_qbz
Requirement already satisfied: contourpy>=1.0.1 in c:\users\sweet\appdata\local\packages\pythonsoftwarefoundation.python.3.12_qbz5n2k
Requirement already satisfied: cycler>=0.10 in c:\users\sweet\appdata\local\packages\pythonsoftwarefoundation.python.3.12_qbz5n2kfra8
Requirement already satisfied: fonttools>=4.22.0 in c:\users\sweet\appdata\local\packages\pythonsoftwarefoundation.python.3.12_qbz5n2
Requirement already satisfied: kiwisolver>=1.3.1 in c:\users\sweet\appdata\local\packages\pythonsoftwarefoundation.python.3.12_qbz5n2
Requirement already satisfied: packaging>=20.0 in c:\users\sweet\appdata\local\packages\pythonsoftwarefoundation.python.3.12_qbz5n2kf
Requirement already satisfied: pillow>=8 in c:\users\sweet\appdata\local\packages\pythonsoftwarefoundation.python.3.12_qbz5n2kfra8p0\
Requirement already satisfied: pyparsing>=3 in c:\users\sweet\appdata\local\packages\pythonsoftwarefoundation.python.3.12_qbz5n2kfra8
Requirement already satisfied: six>=1.5 in c:\users\sweet\appdata\local\packages\pythonsoftwarefoundation.python.3.12_qbz5n2kfra8p0\l
Note: you may need to restart the kernel to use updated packages.

[notice] A new release of pip is available: 25.0.1 -> 26.0.1
[notice] To update, run: C:\Users\sweet\AppData\Local\Microsoft\WindowsApps\PythonSoftwareFoundation.Python.3.12_qbz5n2kfra8p0\python
```

```python
import pandas as pd
from sklearn.linear_model import LinearRegression
import matplotlib.pyplot as plt
import os

def load_and_train():
    if not os.path.exists('study_data.csv'):
        print("❌ Error: 'study_data.csv' not found.")
        return None
    df = pd.read_csv('study_data.csv')
    X = df[['difficulty', 'importance', 'current_knowledge', 'days_to_exam', 'distraction_level']]
    y = df['hours_needed']
    model = LinearRegression()
    model.fit(X.values, y)
    return model

def visualize_plan(user_data, prediction):
    labels = ['Difficulty', 'Importance', 'Knowledge', 'Days Left', 'Distractions']
    values = user_data[0]

    # Create the figure
    plt.figure(figsize=(10, 5))

    # Plot 1: Input Factors
    plt.subplot(1, 2, 1)
    plt.bar(labels, values, color='skyblue')
    plt.title('Your Input Factors')
    plt.ylim(0, 10) # Assuming scale 1-10
    plt.xticks(rotation=45)

    # Plot 2: AI Result
    plt.subplot(1, 2, 2)
    plt.bar(['Suggested Hours'], [prediction], color='salmon')
```

[4] ✓ 9.3s

```python
    plt.subplot(1, 2, 2)
    plt.bar(['Suggested Hours'], [prediction], color='salmon')
    plt.title('AI Prediction')
    plt.ylabel('Hours')

    plt.tight_layout()
    plt.show()

def main():
    ai_model = load_and_train()
    if ai_model:
        print("\n--- 🧠 AI Study Planner with Visualization ---")
        try:
            diff = float(input("Topic Difficulty (1-10): "))
            imp = float(input("Importance (1-10): "))
            know = float(input("Current Knowledge (1-10): "))
            days = float(input("Days until Exam: "))
            dist = float(input("Distraction Level (1-10): "))

            user_input = [[diff, imp, know, days, dist]]
            prediction = ai_model.predict(user_input)[0]
            final_hours = max(0.5, round(prediction, 2))

            print(f"\n✅ AI Recommendation: {final_hours} Hours")

            # Trigger the chart
            print("Generating chart...")
            visualize_plan(user_input, final_hours)

        except ValueError:
            print("❌ Invalid input. Please enter numbers.")

if __name__ == "__main__":
    main()
```

```
--- 🧠 AI Study Planner with Visualization ---

✅ AI Recommendation: 1.59 Hours
Generating chart...
```



AIML.ipynb ● | study_data.csv × | Release Notes: 1.113.0

study_data.csv

```csv
difficulty,importance,current_knowledge,days_to_exam,distraction_level,hours_needed
8,9,2,5,3,14
3,4,8,10,2,2
5,6,5,7,4,6
9,10,1,3,1,18
2,2,9,14,2,1
7,8,4,5,5,11
6,5,6,8,3,7
10,10,2,2,1,20
4,3,7,12,2,3
8,7,3,4,4,12
5,5,5,10,3,5
2,9,8,15,2,2
7,6,4,6,5,10
9,8,2,3,2,16
1,2,10,20,1,0.5
```